



CLIENT APPROVAL PACK

Functional Scope and Approval Document

QR-enabled equipment master, field service capture, signatures, PDF reporting and email distribution

Prepared for	Document	Status	Date
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Version 1.0

PURPOSE: This document converts the approved concept into a clear, phased delivery commitment. It is intended to support stakeholder alignment and formal client approval.

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Document control

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Confidentiality	Confidential - limited to authorized client and delivery stakeholders

APPROVAL EFFECT: Once signed, this document becomes the agreed functional baseline. UI mockups remain design references; this scope and its acceptance criteria govern what will be delivered.

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1. Purpose and approval objective

This document defines the functional and non-functional scope for enhancing the Promach Digital Service Report application into a QR-enabled equipment maintenance platform. It translates the client

concept into implementable requirements, phased deliverables, business rules, acceptance criteria and explicit boundaries.

Approval authorizes detailed design and implementation against this baseline. Items not stated as included are excluded unless added through the change-control process.

1.1 Business objectives

- Minimize technician input and eliminate normal re-entry of equipment master data.
- Ensure the correct checklist and measurements are presented for every equipment visit.
- Preserve equipment-level service history and evidence.
- Allow administrators to change operational configuration without programming changes.
- Produce professional, traceable service reports with configurable acknowledgement and delivery.
- Provide a modern, responsive experience suitable for field phones, tablets and administration desktops.

2. Solution definition and current platform baseline

2.1 Product definition

The proposed product is a browser-based equipment service and digital reporting platform. It will support administrator configuration, technician field capture, client acknowledgement, PDF generation, email distribution and equipment-level history.

2.2 Existing platform capabilities to be retained

Existing capability	Treatment in this scope
Secure administrator login	Retain and extend to technician roles and production access policies.
MongoDB-backed operational records	Retain and extend the data model for QR, hierarchy, schedules, readings, media and email evidence.
Master-data CRUD with pagination	Retain and expand fields, hierarchy and configuration entities.
Guided report creation	Retain for administrative use and add the QR-led technician workflow.
Checklist and measurement capture	Retain and make dynamically configurable by category, frequency and parameter assignment.
Secure client link and admin-device signature	Retain and extend with configurable signature modes.
Locked completed report and audit history	Retain and strengthen revision, grouping and delivery rules.
Branded PDF generation	Retain and extend for category numbering, abnormal readings and optional photos.

2.3 New or materially enhanced capabilities

- Equipment QR generation, printing, scanning, replacement and deactivation.

- Building, level and room/location hierarchy.
- Expanded equipment specifications and lifecycle details.
- Frequency-based checklist selection and effective-dated checklist versions.
- Admin-selected measurement parameters and normal limits.
- Single-phase and three-phase reading layouts.
- Technician-focused mobile service mode with minimal input.
- Equipment service history and advanced signature grouping.
- Category-specific report sequences.
- Photo compression and email recipient/delivery records.

3. Scope principles and delivery phases

GOVERNING PRINCIPLE: Admin configures once; technician scans, completes, records actual results and saves. Master facts and rated values are never re-keyed during a normal service visit.

PHASE 1

Core QR-Enabled Service Operations

End-to-end equipment onboarding, QR scanning, dynamic checklist and measurement capture, service history, baseline acknowledgement, PDF and email.

PHASE 2

Advanced Control, Distribution and Enterprise Readiness

Advanced signature grouping, category numbering, exception reporting, email evidence, revisions, dashboards and production hardening.

3.1 Phase dependency

Phase 2 builds on the master data, transaction records and technician flow established in Phase 1. Phase 1 must pass agreed UAT and pilot exit criteria before Phase 2 production activation.

3.2 Delivery sequencing

1. Scope approval and configuration workshop.
2. Detailed field mapping, wireframes and technical design.
3. Phase 1 build, system testing, UAT and pilot.
4. Phase 1 acceptance and controlled rollout.

5. Phase 2 build, integration testing and UAT.
6. Production hardening, training and final acceptance.

Detailed dates, staffing, commercials and service-level commitments are intentionally excluded from this functional approval document and will be confirmed separately.

4. Users, roles and access

Role	Primary permissions	Restrictions
System Administrator	User access, global settings, all masters, QR, reports, audit and configuration.	Cannot alter signed historical content without a revision process.
Operations Administrator	Client/site/equipment masters, templates, report oversight, QR printing and distribution.	No security-policy administration unless granted.
Technician	Scan QR, view read-only equipment facts, complete assigned service records, readings, remarks and photos.	Cannot create or edit master equipment facts, limits or templates.
Supervisor / Reviewer	Review service results, exceptions, history and release readiness.	Cannot silently edit technician evidence or completed records.
Client Signer	Review the intended service output and sign when required.	Access limited to the secure report/group presented.
Management Viewer	Read dashboards, service history and completed reports.	Read-only.

4.1 Authentication and identity

- Technician name is taken from the authenticated user profile and is not typed for each visit.
- Server date/time is recorded for service events, signatures and email actions.
- Role checks are enforced server-side for every protected operation.
- Inactive users cannot create new service activity.
- Client signing links use high-entropy tokens, can expire or be replaced, and expose only the intended report.

5. Phase 1 detailed functional scope

5.1 Organization and location hierarchy

Admin shall create and maintain the hierarchy used to locate equipment and populate service records.

Entity	Required information	Key rule
Client	Name, address, contacts, email, phone, active status.	A client may contain multiple sites.
Site	Name, address, client, operational contacts, active status.	A site belongs to one client.
Building	Name/code, site, address notes, active status.	A building belongs to one site.
Level	Name/number, building, active status.	A level belongs to one building.
Room / Location	Name/code, level, access notes, active status.	Equipment points to one active room/location.

Deleting a hierarchy record that is referenced by equipment or historical service records shall be blocked. Admin may deactivate it for future use while history remains readable.

5.2 Equipment category master

- Admin can create categories such as FCU, AHU, FAHU, EF, Pump and future categories.
- Each category has a unique short code used in configuration and Phase 2 report numbering.
- Category records can define default checklist, measurement and signature behaviour.
- Inactive categories remain visible in history but cannot be assigned to new equipment.

5.3 Equipment master

Equipment Master is admin-only for creation and maintenance. Technician screens display the applicable fields as read-only.

Field group	Fields delivered	Business rule
Identity	System equipment ID; equipment tag number; equipment name; equipment category; active/inactive status.	Tag number must be unique within the approved client/site rule.
Ownership and location	Client; site; building; level; room/location.	Full hierarchy auto-populates after QR scan.
Manufacturer details	Brand; model; serial number.	Serial validation/uniqueness rule is configurable because some legacy data may be incomplete.
Technical rating	Capacity; refrigerant type; motor type; rated voltage; rated current; rated frequency.	Ratings are never re-entered by technicians.
Lifecycle	Installation date; warranty end date; equipment photo.	Expiry/due indicators may be shown to admin.
Maintenance configuration	Maintenance frequency; required checklist; required measurement parameters; normal min/max limits.	Configuration drives the technician form.
Completion configuration	Signature mode; preset email recipients.	Defaults can be overridden only where permitted.
System controls	QR token/status; creation/update actor and timestamp.	QR token is opaque and not a raw database ID.

5.3.1 Equipment master operations

- Create, view, edit, activate/deactivate, search, filter and paginate equipment records.
- Validate mandatory values and prevent invalid hierarchy combinations.
- Prevent physical deletion when service history exists.
- Show QR status, last service date and next applicable maintenance frequency.
- Allow authorized export of equipment lists and QR label batches if approved in detailed design.

5.4 QR Code Generator

Admin shall have a dedicated QR Code Generator module. One active QR code identifies one equipment record only.

- Generate a unique QR token when equipment is created or on authorized request.
- Preview and download an individual QR image and print-ready label.
- Support batch label generation for selected equipment.
- Display equipment tag, category and concise location text beside the QR code.
- Reissue a damaged or compromised QR code; the previous token becomes invalid.
- Deactivate QR access when the equipment is inactive or retired.
- Record who generated, printed, reissued or deactivated each QR code and when.
- Do not encode confidential master details or predictable database identifiers in the visible QR payload.

PHYSICAL LABEL RESPONSIBILITY: The platform generates print-ready QR labels. Printer selection, label material, physical printing and installation on equipment are client operational responsibilities unless separately included.

5.5 Checklist configuration

Admin shall configure checklist templates without programming changes. Templates are assigned by equipment category and maintenance frequency.

Frequency	Configuration behaviour	Service-time behaviour
Monthly	Admin maintains the monthly template and effective version.	Loaded when the visit is classified as monthly.
Bi-Monthly	Admin maintains a two-month template or rule.	Only applicable items appear.
Quarterly	Admin defines whether quarterly replaces or includes monthly work.	The approved rule is applied automatically.
Half-Yearly	Admin maintains the six-month template and any cumulative rule.	Only the resolved template snapshot is shown.
Yearly	Admin maintains annual maintenance tasks.	Annual checklist and selected parameters load.

5.5.1 Template controls

- Create, edit, activate/deactivate, copy and version templates.
- Add, remove, reorder and mark items mandatory or optional.
- Configure response type where needed: completed/not completed/not applicable, actual reading, text or selection.
- Set effective date and retain prior versions for historical service records.
- Prevent an in-use completed service record from changing when a template is later edited.

- Preview the technician form before activation.

5.6 Measurement parameter and normal-limit configuration

Parameter	Typical units / fields	Display rule
Current	A; single value or R/Y/B phase values.	Shown only when selected; phase layout follows motor type.
Voltage	V; single value or phase-to-phase/phase values.	Shown only when selected.
Temperature	deg C.	Shown only when selected; normal min/max can be configured.
Relative humidity	% RH.	Shown only when selected.
Pressure	Pa, kPa, bar or configured unit.	Unit is selected by admin.
Airflow	CFM, m3/h or configured unit.	Unit is selected by admin.
Frequency	Hz.	Rated frequency comes from Equipment Master; actual reading is entered if required.
Custom parameter	Admin-defined label and unit.	Available without programming changes within supported field types.

- Admin selects which parameters apply to a category/template/equipment.
- Admin configures normal minimum and maximum values where applicable.
- Technician enters only the actual reading.
- Rated values remain visible as read-only reference values.
- Parameters not selected by admin do not appear in the technician workflow or corresponding PDF section.
- Each completed service stores a snapshot of labels, units, limits and readings.

5.7 Motor phase behaviour

Configured motor type	Technician fields	Rule
Single Phase	Only the approved single-phase current/voltage fields.	Three-phase fields remain hidden.
Three Phase	Only the approved R/Y/B or phase-pair readings.	Field naming is confirmed during design.
Not Applicable	No motor-phase reading group.	Only other selected parameters appear.

5.8 Technician scan-to-service workflow

01 SIGN IN	02 SCAN QR	03 COMPLETE	04 SAVE	05 SCAN NEXT
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5.8.1 Automatic population

- Equipment identity, client and location hierarchy.

- Category, brand, model, serial and technical ratings.
- Logged-in technician identity.
- Server date/time and service session identifier.
- Applicable checklist and frequency.
- Selected measurement parameters, units and normal limits.
- Configured signature and email defaults.

5.8.2 Technician-entered information

- Checklist responses.
- Actual measurement readings.
- Abnormal status only when required.
- Free-text remarks when necessary.
- Optional photos when necessary.
- Completion/save action.

5.8.3 Technician restrictions

- Technician cannot edit equipment master facts, ratings, normal limits or template definitions.
- Technician cannot manually select another equipment while completing a QR-opened service record.
- Any emergency manual lookup, if approved, is permission-controlled and audited.
- Invalid, expired, replaced or inactive QR codes do not open an editable service form.

5.9 Abnormal-reading interaction

- Default reading status is Normal.
- There are no permanent High / Low / Normal button groups.
- Technician can tap or long-press a reading to mark it abnormal and the reading is highlighted red.
- An accessible visible action is also provided so the function is discoverable and usable without long-press.
- When limits exist, the system compares the actual reading and warns when it is outside the configured range.

- The recommended rule requires technician confirmation before completing an out-of-range reading as Normal.
- Abnormal status, actual value, configured limits, technician and timestamp are stored.

CLIENT DECISION: Confirm whether out-of-limit readings must be automatically forced to Abnormal or only warned for technician confirmation.

5.10 Service session, save and equipment history

- A service session begins when a valid equipment QR is opened.
- Draft progress is autosaved where technically feasible and can be resumed by the authorized technician.
- Mandatory checklist items and required readings are validated before completion.
- Completion creates an immutable equipment service-history entry or controlled revision.
- History shows service date/time, technician, frequency, checklist results, readings, exceptions, remarks, photos, signature linkage and report linkage.
- Search and filters support equipment tag, category, client, site, date, technician and status.
- Historical records retain the master and template snapshot used at the time.

5.11 Remarks and photos

- Remarks are optional free text unless an abnormal item rule makes a remark mandatory.
- Photos are optional and may be captured from the device camera or selected from supported files.
- The application compresses images while preserving useful service evidence.
- File type, size and content-type validation is enforced.
- The system targets an email package below 10 MB; if the final package remains larger, it uses the approved secure-link fallback or another agreed rule.
- AI recommendations are not included; the data model may preserve a future extension point.

5.12 Phase 1 acknowledgement, PDF and email

Phase 1 includes a complete end-to-end output using the existing overall-report signature or no-client-signature email-only mode. Advanced grouping modes are delivered in Phase 2.

- Secure client link or admin/technician device signature for the overall report.

- Email-only completion where the configured equipment/report requires no client signature.
- Locked completed record with signer identity, designation, time, channel and consent where signed.
- Professional PDF containing applicable equipment, checklist, selected measurements, exceptions, remarks and optional photos.
- Preset recipients with authorized manual additions.
- Sent date, time and recipient list recorded against the report.

5.13 Phase 1 acceptance gate

PHASE 1 EXIT CRITERIA: A client-approved pilot demonstrates equipment creation, QR label generation, successful mobile scan, automatic form loading, technician completion, history creation, acknowledgement or email-only completion, PDF generation and controlled email distribution.

6. Phase 2 detailed functional scope

6.1 Advanced signature modes

Signature mode	System behaviour	Result
Individual equipment signature	Signature is collected for each completed equipment item.	Each equipment history entry carries its own acknowledgement.
Selected equipment completed today	Authorized user selects eligible equipment from the same client/date/service context.	One signature covers the recorded selection.
Equipment category signature	Completed equipment is grouped by category.	One signature covers the category group.
Overall service report	All eligible equipment in the report is grouped.	One signature covers the complete report.
No client signature	Report is completed under email-only rule.	Email evidence replaces client acknowledgement; record clearly states no signature required.

6.1.1 Signature controls

- Signature mode defaults from Equipment Master or category configuration.
- A grouping cannot combine equipment from different clients.
- The covered equipment IDs and service revisions are frozen at signing.
- The signer sees what is being acknowledged before signing.
- Signed content is locked and any correction creates a traceable revision.
- Signature image, signer, designation, email, timestamp, channel, consent and covered records are retained.

6.2 Category-specific report numbering

The platform shall maintain a separate concurrency-safe sequence for every configured equipment category.

Element	Rule	Example
Category prefix	Taken from the unique equipment category code.	FCU, AHU, EF
Date segment	Generated from the approved service/report date using YYYY-MMDD unless changed during design.	2026-0728
Sequence	Four-digit category sequence with atomic allocation.	0001
Complete number	Prefix + date segment + sequence.	FCU-2026-0728-0001
Reset policy	Client must approve daily, yearly or continuous reset.	Decision required
Cancellation	Allocated numbers are not silently reused.	Audit retains cancelled number

6.3 Advanced PDF behaviour

- Generate one or more PDFs according to the approved signature and category grouping.
- Show only parameters selected for the serviced equipment.

- Highlight abnormal readings in red and include an exception summary.
- Include normal readings or suppress them according to the approved PDF decision.
- Include optional photos with compression and captions where provided.
- Display report number, equipment identity, location, technician, service date/time, checklist version, signature evidence and issue metadata.
- Preserve a generated-document version and checksum for the signed/issued copy.
- Prevent silent regeneration of a different PDF under the same completed version.

6.4 Email recipient and delivery management

- Admin configures multiple preset recipients at client, site, category or equipment level.
- Authorized sender may add validated manual recipients before sending.
- One-click Send Report creates a delivery event.
- System records sent date/time, sender, To/CC recipients, report/document version and delivery status.
- Failed sends are visible and can be retried without creating a false completion record.
- Recipient addresses are validated and duplicates are removed.
- Large packages follow the approved compression or secure-link fallback rule.
- Email sender domain, templates and provider configuration are confirmed before production.

6.5 Correction and revision control

- Draft and in-progress records remain editable by authorized roles.
- Signed, email-only completed or issued records are locked.
- Correction creates a new linked revision and preserves the prior record and PDF.
- Audit history identifies the reason, requester, actor, time and changed revision.
- A corrected report follows the applicable signature and delivery process again.

6.6 Dashboards and operational views

- Report counts by draft, in progress, awaiting signature, completed, correction required and cancelled.
- Equipment service history and latest service status.
- Abnormal reading and follow-up indicators.

- Email sent/failed status and signature status.
- Search, filtering, pagination and export where approved.
- Master-data coverage and inactive/missing-configuration indicators.

6.7 Production readiness

- Operational monitoring and error logging.
- Database backup and tested restore procedure.
- Secure secret and environment configuration.
- File/media storage controls and retention.
- Security testing and remediation of critical/high findings.
- User guides, admin configuration guide and technician quick-start guide.
- Pilot support, defect triage, go-live checklist and rollback plan.

6.8 Phase 2 acceptance gate

PHASE 2 EXIT CRITERIA: All approved signature modes, category sequences, advanced PDF rules, recipient management, delivery evidence, revision controls and operational readiness tests pass client UAT.

7. Cross-cutting business rules

Rule ID	Rule
BR-01	One active QR code identifies one equipment record only.
BR-02	A replaced QR token is invalid immediately after replacement.
BR-03	Technicians cannot edit equipment master information during service.
BR-04	Checklist and parameter display is controlled by active approved configuration.
BR-05	Completed history stores a snapshot and is not changed by later master edits.
BR-06	Only selected measurement parameters are displayed and reported.
BR-07	Motor reading fields follow the equipment motor type.
BR-08	Reading status defaults to Normal until marked or confirmed abnormal.
BR-09	Out-of-limit handling follows the client-approved enforcement rule.
BR-10	Abnormal items retain actual value, limits, remark where required and actor/time.
BR-11	A signature covers only the explicitly presented equipment/service revisions.
BR-12	Completed or issued records are locked; correction uses revision control.
BR-13	Report numbers are allocated atomically and are not silently reused.
BR-14	Email events retain sender, recipients, document version, time and result.
BR-15	Historical records remain visible when a master record is deactivated.
BR-16	Manual equipment lookup, if approved, is exceptional and audited.
BR-17	Client, site and equipment isolation is enforced server-side.
BR-18	Physical QR label printing and fixing is client responsibility unless separately contracted.

8. Data and record model

Domain	Primary records	Important relationships
Identity	Users, roles, sessions.	User actions link to audit events and service records.
Organization	Clients, contacts, sites, buildings, levels, rooms.	Equipment belongs to one valid location hierarchy.
Equipment	Categories, equipment, specifications, photos, QR tokens.	QR resolves to one active equipment record.
Configuration	Checklist templates/versions/items, parameters, units, limits, frequency rules, signature rules, recipients.	Active configuration resolves the technician form.
Service	Service sessions, equipment visits, checklist results, readings, exceptions, remarks, media.	Each visit belongs to equipment and a configuration snapshot.
Completion	Signature groups, acknowledgements, revisions, report numbers, generated PDFs.	Completion freezes the covered service revisions.
Distribution	Email messages, recipients, delivery attempts and outcomes.	Every send references the issued document version.
Governance	Audit events, status transitions, access and configuration history.	Sensitive actions are attributable and timestamped.

8.1 Snapshot and history rule

Every completed equipment visit stores the equipment identity, location, category, checklist version, parameter definitions, limits, readings and technician identity used at that moment. A later master change affects future visits only.

8.2 Deletion and deactivation

Referenced master records are deactivated rather than physically deleted. Unreferenced draft/test records may be deleted by authorized admin according to the approved policy.

9. User experience and device requirements

Area	Requirement
Mobile field use	Responsive at 360 px and above; single-column flow; no horizontal scrolling for core tasks.
Tap targets	Frequent controls target at least approximately 44 px.
QR scanning	Camera permission guidance, retry, torch support where browser permits, and clear invalid-code messages.
Progress	Visible equipment identity, service progress, save state and Scan Next action.
Input minimization	No manual master data entry; numeric keyboard for readings; sensible defaults.
Accessibility	Labels, focus order, keyboard support, contrast and non-long-press alternative for abnormal marking.
Desktop administration	Persistent navigation, dense but readable tables, search, filters and pagination.
Supported browsers	Current production versions of Chrome/Edge and mobile Safari/Chrome; exact matrix confirmed before UAT.

9.1 Offline and GPS boundary

The approved scope assumes an active network connection. Full offline synchronization, GPS attendance and geofencing are excluded unless separately scoped. The auto-populated location is the configured equipment location, not device GPS.

10. Security, privacy and audit requirements

- TLS for production traffic and secure session cookies.
- Role-based and tenant-aware authorization on the server.
- High-entropy QR and client-link tokens; only hashes stored where appropriate.
- Rate limiting for authentication and public token endpoints.
- Input validation and injection protection.
- File type, size and content checks for uploaded photos.
- Secrets stored outside source control.
- Audit events for master changes, QR actions, service completion, signatures, revisions, PDF generation and emails.
- User and recipient personal data limited to operational purpose.
- Retention, deletion and hosting-region policy confirmed by client before production.

10.1 Audit event minimum fields

Each sensitive event shall store event type, target record, actor or token channel, server timestamp, relevant status transition and a human-readable description. Audit records must not expose passwords, secrets or full QR tokens.

11. Performance, reliability and operational requirements

NFR ID	Requirement	Acceptance approach
NFR-01	QR scan should open the equipment form promptly on an adequate mobile connection.	Target and test network agreed before UAT; proposed P95 within 3 seconds excluding first camera permission.
NFR-02	Normal admin page actions should feel responsive.	Proposed P95 server response within 2.5 seconds excluding uploads/PDF/email provider latency.
NFR-03	Draft service progress shall survive normal navigation/reload according to autosave design.	Interrupt-and-resume UAT scenario.
NFR-04	Concurrent report-number allocation shall not produce duplicates.	Concurrency integration test.
NFR-05	Photo upload and compression shall not freeze the field workflow.	Representative phone/photo test set.
NFR-06	Generated PDFs shall be readable, complete and version-linked.	Content and visual regression test.
NFR-07	Backups and restore procedure shall be documented and tested before production.	Recorded restore rehearsal.

NFR ID	Requirement	Acceptance approach
NFR-08	No open critical or high-severity defect at production approval.	Defect and security review.

Final availability, recovery objectives, volume limits and support service levels require separate commercial/operational confirmation.

12. Reporting, PDF and email specification

12.1 PDF minimum content

- Promach branding and approved company details.
- Report number, status and generated version.
- Client and full service location.
- Equipment category, tag, name, brand, model and serial.
- Service date/time, frequency and technician.
- Applicable checklist results.
- Only the selected measurement parameters.
- Abnormal readings highlighted and summarized.
- Remarks and approved optional photos.
- Signature/acknowledgement evidence or clear no-signature status.
- Page numbers and document issue metadata.

12.2 PDF measurement decision

RECOMMENDED RULE: Show every selected actual reading and visually highlight abnormal values. Also provide an exception-only summary. This preserves a complete maintenance record while making abnormalities obvious.

If the client instead approves an exception-only PDF, normal readings will remain in the system history but will be omitted from the client PDF.

12.3 Email record

- Subject and template identify the report/equipment grouping.
- Preset and manual recipients are visible before sending.
- Issued PDF or approved secure link is attached/included.

- Successful and failed delivery attempts are logged.
- Sent record includes sender, recipients, time and document version.

13. Testing, UAT and acceptance

13.1 Test coverage

- Unit tests for validation, checklist resolution, limits, numbering and status rules.
- Integration tests for database, QR resolution, media, PDF and email provider interactions.
- Authorization tests for roles, client isolation, inactive records and guessed IDs/tokens.
- End-to-end tests for equipment onboarding, scan, service, sign, report and email.
- Responsive tests for representative phone, tablet and desktop viewports.
- Accessibility checks for labels, focus, keyboard operation, contrast and abnormal-marking alternative.
- Security checks for sessions, uploads, rate limits and sensitive data leakage.
- Visual verification of generated PDFs.

13.2 UAT process

1. Client nominates UAT users and representative equipment/categories.
2. Approved master data and checklist samples are loaded.
3. Delivery team demonstrates the agreed end-to-end scenarios.
4. Client records defects against agreed acceptance criteria.
5. Critical/high defects are resolved; lower-severity disposition is agreed.
6. Client signs the phase acceptance record.

13.3 Phase 1 acceptance scenarios

ID	Scenario	Expected result
P1-AC-01	Admin creates full location hierarchy and equipment with required fields.	Saved record validates and appears in Equipment Master.
P1-AC-02	Admin generates one unique QR for the equipment.	QR preview/label is printable and resolves only to that equipment.
P1-AC-03	Technician scans the valid QR.	Correct equipment form opens; no manual equipment keying is required.
P1-AC-04	Technician identity, date/time and location populate automatically.	Values match login, server time and equipment master.
P1-AC-05	Monthly/other configured checklist resolves.	Only applicable checklist items appear.
P1-AC-06	Selected measurement parameters resolve.	Only configured fields and phase layout appear.

ID	Scenario	Expected result
P1-AC-07	Technician enters actual readings.	Rated values and limits remain read-only.
P1-AC-08	Technician marks an abnormal reading.	Reading displays red and stores exception evidence.
P1-AC-09	Technician adds optional remarks/photo.	Record saves and image is compressed/validated.
P1-AC-10	Technician saves and scans next.	First visit remains in equipment history.
P1-AC-11	Overall acknowledgement or email-only rule is completed.	Record locks under correct completion rule.
P1-AC-12	PDF is generated and sent.	Applicable content is correct and delivery event is recorded.

13.4 Phase 2 acceptance scenarios

ID	Scenario	Expected result
P2-AC-01	Each signature mode is configured and executed.	Signature covers exactly the intended equipment/service revisions.
P2-AC-02	Concurrent category reports are created.	Unique category numbers are allocated without duplication.
P2-AC-03	Advanced PDF is generated.	Only selected parameters appear; abnormal values follow approved rule.
P2-AC-04	Multiple preset and manual recipients are used.	Validated recipients and sent time are recorded.
P2-AC-05	Email provider returns a failure.	Failure is visible and retry is controlled/audited.
P2-AC-06	Correction is requested after completion.	New revision is created; prior record/PDF remains intact.
P2-AC-07	Dashboard filters service history and exceptions.	Results match stored records and pagination.
P2-AC-08	Backup restore and go-live checks are executed.	Evidence is recorded and client approves production readiness.

14. Deliverables, responsibilities and dependencies

14.1 Delivery-team deliverables

- Approved detailed UX flows and responsive screens.
- Implemented web application modules defined in this scope.
- Database migration and configuration model.
- QR label templates and scanning workflow.
- Configured PDF template and email templates.
- Automated tests and UAT support.
- Deployment/configuration documentation.
- Administrator guide, technician quick-start guide and handover.

14.2 Client responsibilities

- Approve scope, open decisions, branding, wording and acceptance criteria.
- Provide accurate client/site/building/level/room and equipment data.

- Provide checklist content by category and frequency, including mandatory/optional rules.
- Provide measurement definitions, units and normal limits.
- Confirm signature modes, report numbering reset and PDF rule.
- Provide authorized email sender/domain and recipient rules.
- Supply representative phones/tablets, QR printer/labels and UAT users.
- Print and physically affix QR labels unless otherwise contracted.
- Approve retention, hosting, backup and production access policies.
- Complete UAT and provide timely consolidated feedback.

14.3 Dependencies

Dependency	Owner	Impact if unavailable
Approved equipment data and hierarchy	Client	QR and pilot records cannot be validated.
Approved checklist/frequency mapping	Client	Dynamic form behaviour cannot be finalized.
Approved limits and units	Client	Abnormal detection cannot be validated.
Camera-capable supported devices	Client	Field QR UAT cannot be completed.
Email provider and sender domain	Joint	Production email delivery cannot be activated.
Hosting/database/media environments	Joint	Production readiness and operating targets remain pending.
Timely UAT feedback	Client	Phase acceptance and rollout are delayed.

15. Assumptions, exclusions and change control

15.1 Assumptions

- The solution is delivered as a responsive web application requiring network connectivity.
- Equipment data and maintenance rules supplied by the client are accurate and approved.
- One equipment belongs to one active client/location hierarchy at a time.
- Standard phone cameras can read the printed QR labels under normal site conditions.
- Photo compression is best effort and secure-link fallback is acceptable when needed.
- Commercial schedule, support SLA and infrastructure costs are governed separately.

15.2 Explicit exclusions

- AI-generated recommendations or automated diagnosis.
- Offline synchronization.
- GPS tracking, geofencing, attendance or payroll.

- Preventive-maintenance job scheduling and dispatch optimization.
- Inventory, spare parts and purchase management.
- Contracts, quotations, pricing, GST, invoicing or payments.
- IoT/sensor integration and automatic reading capture.
- OCR or migration from handwritten service reports.
- WhatsApp/SMS notifications.
- Native iOS/Android applications.
- Custom QR printing hardware, labels or physical installation.
- Third-party system integrations not explicitly listed.

15.3 Change control

A request that changes approved fields, workflows, roles, integrations, reports, non-functional targets or acceptance criteria will be documented as a change request. The delivery team will assess scope, schedule, commercial and technical impact before implementation. No change is effective until authorized by the nominated client and delivery representatives.

16. Open decisions requiring client confirmation

ID	Decision	Recommended position	Client response
D-01	PDF reading display	All selected readings with abnormal highlight and exception summary.	
D-02	Frequency composition	Admin-configurable replacement or cumulative rule.	
D-03	Out-of-limit enforcement	Automatic warning plus technician confirmation; consider forced abnormal.	
D-04	QR label format	Confirm label dimensions, included text and batch layout.	
D-05	Damaged QR fallback	Audited admin-authorized lookup only.	
D-06	Report number reset	Select daily, yearly or continuous per category.	
D-07	Signature grouping precedence	Confirm equipment override versus category/client default.	
D-08	Email package over 10 MB	Use secure download link after compression threshold.	
D-09	Email sender	Confirm provider, sender domain and reply-to mailbox.	
D-10	Retention	Confirm service, audit, photo, PDF and email retention period.	
D-11	Hosting and data region	Confirm production region, access and backup requirements.	
D-12	Supported device/browser list	Confirm representative field devices for UAT.	

17. Approval and sign-off

By signing below, the parties confirm that this document accurately describes the functional scope proposed for the Promach QR Equipment Service Platform. Approval includes the two-phase structure, requirements, acceptance approach, responsibilities, assumptions and exclusions, subject to any written decision responses attached to this document.

17.1 Approval selection

- ☐ Approve both Phase 1 and Phase 2 as scoped.
- ☐ Approve Phase 1 only; Phase 2 remains subject to later authorization.
- ☐ Approve subject to the written clarifications attached.
- ☐ Return for revision.

17.2 Client approval

Field	Client response
Organization	PROMACH PTE. LTD.
Name	
Designation	
Signature	
Date	
Comments / conditions	

17.3 Solution delivery acknowledgement

Field	Delivery-team response
Organization	Solution Delivery Team
Name	
Designation	
Signature	
Date	

Appendix A. Functional requirement register

ID	Actor	Requirement	Phase	Priority
FR-001	Admin	Manage client/site/building/level/room hierarchy.	P1	Must
FR-002	Admin	Manage equipment categories and unique category codes.	P1	Must
FR-003	Admin	Manage all approved Equipment Master fields.	P1	Must
FR-004	Admin	Generate one unique active QR per equipment.	P1	Must
FR-005	Admin	Preview, download, print, batch and reissue QR labels.	P1	Must
FR-006	Admin	Configure checklist templates without code changes.	P1	Must
FR-007	Admin	Version checklist templates by frequency/effective date.	P1	Must
FR-008	Admin	Configure parameters, units and normal limits.	P1	Must
FR-009	System	Resolve QR to one active equipment.	P1	Must
FR-010	System	Auto-populate equipment, technician, time and location.	P1	Must
FR-011	System	Load only applicable checklist items.	P1	Must
FR-012	System	Load only selected measurement parameters.	P1	Must
FR-013	System	Display fields according to single/three-phase configuration.	P1	Must
FR-014	Technician	Enter actual readings only.	P1	Must
FR-015	Technician	Mark abnormal by tap/long-press accessible interaction.	P1	Must
FR-016	Technician	Add optional remarks and compressed photos.	P1	Must
FR-017	Technician	Save and scan next equipment.	P1	Must
FR-018	System	Store equipment-level service history.	P1	Must
FR-019	System	Support overall signature or email-only completion.	P1	Must
FR-020	System	Generate professional PDF and send by email.	P1	Must
FR-021	Admin	Configure all five signature modes.	P2	Must
FR-022	System	Group covered equipment safely for signature.	P2	Must
FR-023	System	Allocate category-specific running report numbers.	P2	Must
FR-024	System	Apply advanced PDF abnormal-reading rule.	P2	Must
FR-025	Admin	Manage multiple preset recipients.	P2	Must
FR-026	Authorized User	Add validated manual recipients.	P2	Must
FR-027	System	Record send time, sender, recipients and result.	P2	Must
FR-028	System	Lock issued reports and create controlled revisions.	P2	Must
FR-029	Management	View dashboards, history, exceptions and delivery state.	P2	Should
FR-030	Operations	Monitor, back up, restore and support production.	P2	Must

Appendix B. Glossary

Term	Definition
Abnormal reading	A reading explicitly identified as outside normal operation or confirmed as an exception.
Checklist snapshot	The exact checklist version stored with a completed service record.
Equipment Master	The admin-controlled source of equipment identity, location, specifications and service configuration.
Frequency	The maintenance interval used to resolve the applicable checklist, such as Monthly or Quarterly.
Normal limits	Admin-configured minimum and maximum values used to assess an actual reading.
QR token	An opaque, unique value that securely resolves to one equipment record.
Service session	The field activity opened for an equipment item after a valid QR scan.
Signature group	The exact set of equipment service revisions covered by one acknowledgement.
Snapshot	A frozen copy of master/configuration information used in a historical service record.
UAT	User Acceptance Testing performed by nominated client users against approved criteria.

END OF DOCUMENT: Client approval should include responses to the open decisions or an attached clarification record.